# Russell Newton

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## Education

**Georgia Institute of Technology** — Atlanta, GA

**Master of Science in Architecture** — August 2024 to May 2025
GPA: 4.0/4.0
Concentration: Design Computation

**Master of Science in Computer Science** — January 2023 to May 2024
GPA: 4.0/4.0
Concentration: Computer Graphics

**Bachelor of Science in Computer Science** — August 2020 to December 2022
Graduated with Full Honors (GPA: 3.87/4.0)
Concentrations: Artificial Intelligence and Modeling & Simulations

## Experience

**Delta Air Lines** — Atlanta, GA

**IT Associate Developer** — September 2025 to Present

- Implementing new full-stack business requests and maintaining existing features in Delta's flight attendant payroll system.
- Optimizing SQL queries and Java code to improve application responsiveness.
- Modernizing and documenting existing code with an emphasis on performance and readability.

**Norstella** — Atlanta, GA

**Software Engineer Intern** — May 2025 to August 2025

- Implemented new features for Norstella's fine-grained entitlements system to support internal and external clients.
- Led the development of a full-stack MVP designed to present customer-catered pharmaceutical landscape analytics, integrated with the entitlements system enable analysts to control data availability across customer groups, user roles, and drug portfolios.

**Georgia Institute of Technology** — Atlanta, GA

**Graduate Instructor** (ARCH 4833/7030/8833: Design Scripting) — January 2025 to May 2025

- Created course curriculum from personal experience and previous course curriculums.
- Introduced architecture and design students to the uses of Python inside Grasshopper 3D for McNeel's Rhino 8.
- Published the course content and assignments on GitHub Pages.

**Graduate Research Assistant, Shape Computation Lab** — August 2024 to May 2025

- Led development on Shape Machine and the website for the project.
- Rewrote legacy code with modern design patterns and principles.
- Optimized novel computational geometry problems.
- Improved user and developer workflows with new features and processes.

**ScienceLogic** — Reston, VA

**Software Engineering Intern** — June 2022 to December 2022

- Restructured, improved, and tested old code to improve product longevity.
- Developed utilities to easily switch between Python interpreters during system operation.
- Praised by team for ability to work well under pressure and to learn and adapt quickly.

## Project Spotlight

**Shape Machine** | Website
Shape Machine leverages the vector-based representations of geometry within a CAD environment to search for shapes within a design and replace them with new ones. Shape Machine can interpret DrawScript, its very own Turing-Complete programming language where shapes become code.

**Project Abyss (Concept Game)**
A co-op arena shooter created as a concept pitch for the Savanna College of Art and Design VGDev Senior Capstone.

- Contributions: enemy AI, weapon logic, and network-replicated movement.
- Advised the programming of the game based on the pitch, Mind the Abyss (Steam Page).

**MultiTouch** | GitHub | Website
A no-OS gesture recognition library in C for L3Harris as part of the GT CS capstone course. Uniquely, MultiTouch supports inputs from multiple hands at the same time. The website includes a mobile-friendly tech demo.

**TikTokPy** | GitHub | PyPI | ReadTheDocs
A TikTok data scraping library built with Playwright and Pydantic. Built with feedback and contributions from users.

## Skills

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|---|---|
| **Languages:** | Python, C/C++, JavaScript, C#, Java, GLSL, HLSL, SQL |
| **Frameworks & Tools:** | Unreal Engine, Unity, React, Node.js, Windows, Linux, Computer-Aided Design |
| **Applications:** | Multithreading/Concurrency/Parallelism, Computer Graphics, Computational Geometry, Computer Vision, Machine Learning, Full-Stack Development |
| **Automation & CI/CD:** | Docker, GitHub Actions, Jenkins, Playwright |
| **Mathematics:** | Linear Algebra, Multivariable Calculus, Differential Equations, Statistical Modeling |

## Publications

R. Newton and A. Economou, "Redefining Line Maximization," in *Design Computing and Cognition'24*, vol. 1, Springer, Cham, Sep. 2024, pp. 250–264. doi: 10.1007/978-3-031-71918-9_16.

A. Economou, T.-C. K. Hong, and R. Newton, "Shape Meets Euclid: Integrating Shape Computation With Ruler and Compass Procedures," *Automation in Construction*, vol. 165, p. 105562, Jun. 2024, doi: 10.1016/j.autcon.2024.105562.

## Achievements

**Eagle Scout** | Boy Scout Troop 795 July 2019

- Conceptualized a personal community service project with the National Park Service to remove and replace a foot bridge at a local hiking trail.
- Conducted research and designed the new foot bridge, accounting for material properties and site conditions that impact longevity.
- Raised funds for material costs and led several volunteers in the execution of the project.